

# LUXURY FASHION RESALE DATABASE

32606 DATABASE | ASSESSMENT 5 | 2026

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# About Database

This dataset is inspired by the website <https://www.resee.com/en/>

## What is RESEE?

ReSee is a Paris-based luxury pre-loved fashion resale platform, founded in 2013.

The company specialises in buying, authenticating, and reselling high-end designer pieces from iconic brands such as Chanel, Hermès, Celine, Louis Vuitton and Saint Laurent.

Unlike typical resale marketplaces, ReSee offers a full-service experience, where sellers simply submit their pre-loved items and ReSee handles everything: collection, professional photography, authentication, storage, and worldwide shipping to buyers in over 100 countries.





# About Database

This database is a simplified model of how Luxury Fashion Resale platform works. It tracks:

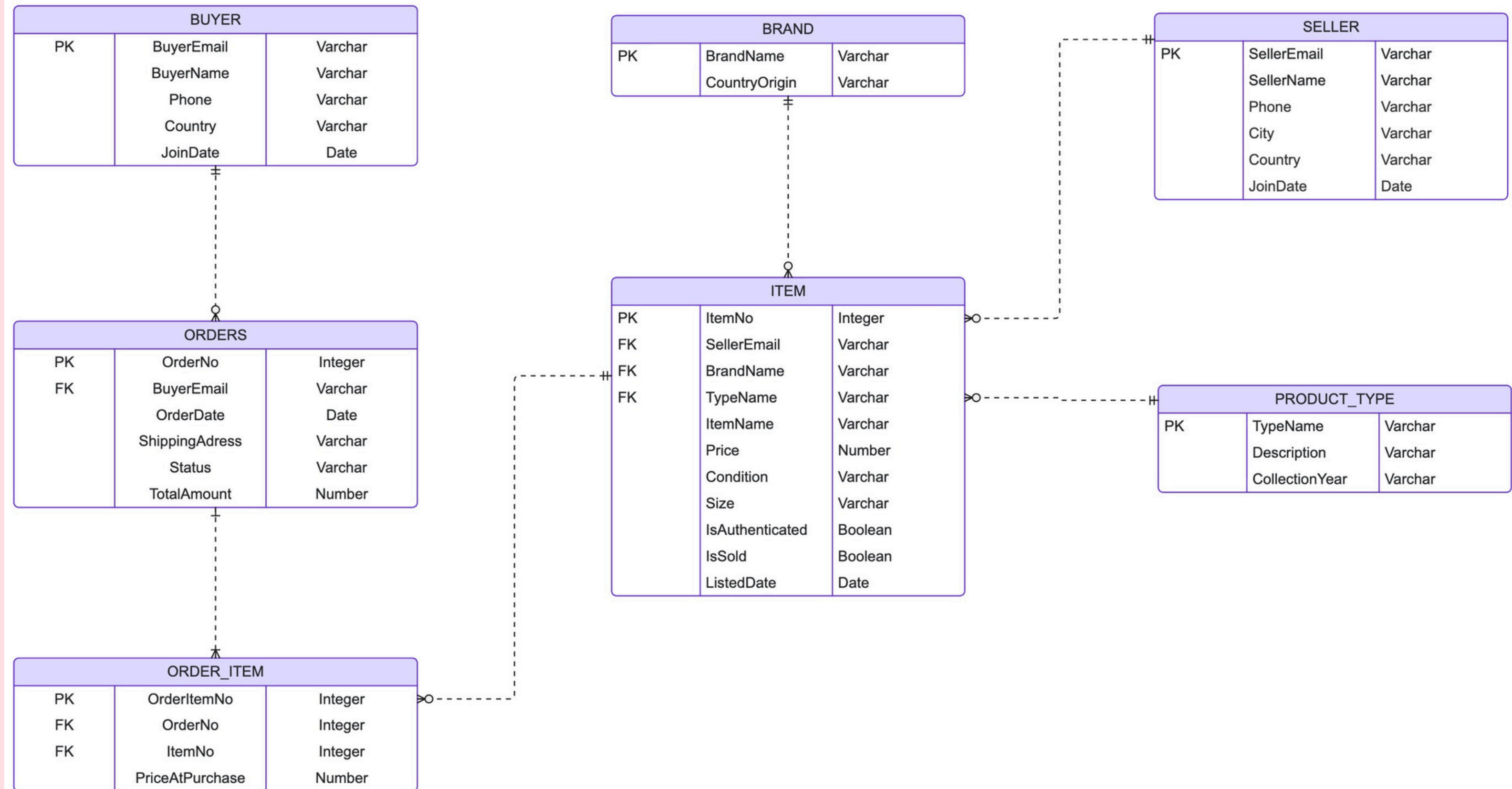
- **Sellers** who submit their designer pieces
- **Buyers** who purchase authenticated items
- **Brands** and their country of origin
- **Product Types** to categorise items (Handbags, Clothing, Shoes, etc.)
- **Items** listed for sale with condition, price and authentication status
- **Orders** placed by buyers
- **Order Items** linking each order to the specific items purchased

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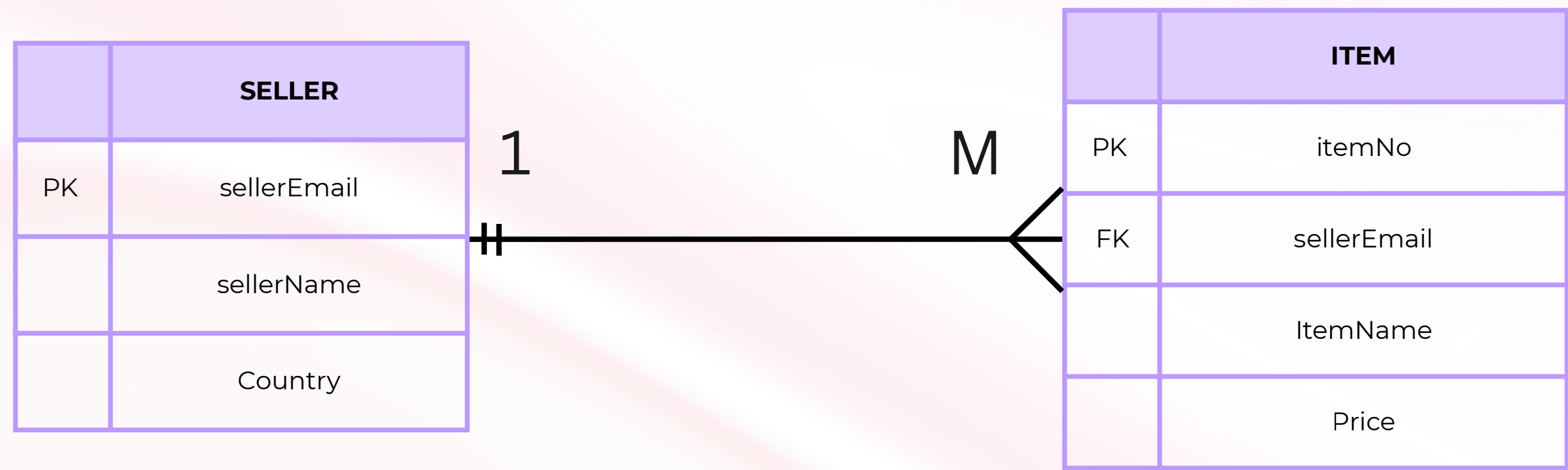
# Entity Relationship Diagram (ERD)





# One-to-Many Relationship: SELLER → ITEM

One seller can submit many items, but each item belongs to exactly one seller.



**EXAMPLE:** Anna Kowalski has 3 items, this demonstrates the one-to-many relationship, 1 seller can send many items..

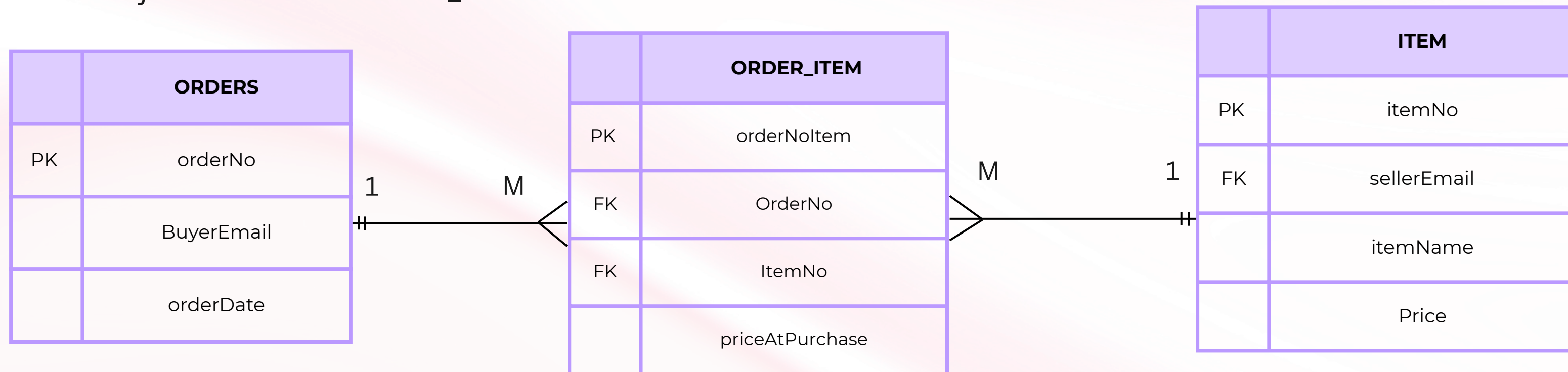
| sellerEmail       | sellerName    | Country |
|-------------------|---------------|---------|
| sofia.b@email.com | Sofia Bernard | France  |
| anna.k@email.com  | Anna Kowalski | UK      |

| ItemNo | sellerEmail       | itemName                |
|--------|-------------------|-------------------------|
| 1      | sofia.b@email.com | Chanel Classic Flap Bag |
| 2      | anna.k@email.com  | Hermes Silk Scarf       |
| 3      | anna.k@email.com  | Celine Luggage Tote     |
| 9      | anna.k@email.com  | Hermes Cape Cod Watch   |



# Many-to-Many: ORDERS ↔ ITEM via ORDER\_ITEM

An order can contain many items, and an item can appear in many orders. This is resolved by the junction table ORDER\_ITEM.



**EXAMPLE:** Order 1004 contains 2 items (3 and 8). Item 3 could appear in other orders too, like order 1006. This is the many-to-many shown by ORDER\_ITEM.

| orderNo | buyerEmail        |
|---------|-------------------|
| 1004    | yuki.m@email.com  |
| 1005    | alice.t@email.com |
| 1006    | emma.b@email.com  |

| orderno | itemNo | priceAtPurchase |
|---------|--------|-----------------|
| 1004    | 3      | 2800.00         |
| 1004    | 8      | 380.00          |
| 1005    | 4      | 1200.00         |
| 1006    | 3      | 2800.00         |

| itemNo | itemName            |
|--------|---------------------|
| 3      | Celine luggage tote |
| 4      | Balenciaga Blazer   |
| 8      | Chanel CC Earrings  |



# Query 1: Simple Single-Table Query

**QUERY 1 :** List the item name, brand, price and condition of all items that are authenticated and still available for sale, ordered by price from highest to lowest.

| SQL Query                                                                                                                                 | Query Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |           |       |           |                               |        |         |           |                                |        |         |           |                          |       |         |           |                             |       |         |           |                           |            |         |      |                               |        |        |           |                          |        |        |           |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-------|-----------|-------------------------------|--------|---------|-----------|--------------------------------|--------|---------|-----------|--------------------------|-------|---------|-----------|-----------------------------|-------|---------|-----------|---------------------------|------------|---------|------|-------------------------------|--------|--------|-----------|--------------------------|--------|--------|-----------|
| <pre>SELECT itemName, brandName, price, condition FROM ReSee_Item WHERE isSold = 'N' AND isAuthenticated = 'Y' ORDER BY price DESC;</pre> | <table><tr><th>itemName</th><th>brandname</th><th>price</th><th>condition</th></tr><tr><td>Chanel Classic Flap Bag Black</td><td>Chanel</td><td>3500.00</td><td>Excellent</td></tr><tr><td>Celine Luggage Tote Phoebe Era</td><td>Celine</td><td>2800.00</td><td>Very Good</td></tr><tr><td>Prada Galleria Bag Black</td><td>Prada</td><td>1800.00</td><td>Excellent</td></tr><tr><td>Gucci Velvet Jacket Size 40</td><td>Gucci</td><td>1500.00</td><td>Very Good</td></tr><tr><td>Balenciaga Blazer Size 38</td><td>Balenciaga</td><td>1200.00</td><td>Good</td></tr><tr><td>Hermes Silk Scarf Les Chevaux</td><td>Hermes</td><td>450.00</td><td>Very Good</td></tr><tr><td>Chanel CC Pearl Earrings</td><td>Chanel</td><td>380.00</td><td>Excellent</td></tr></table> <p>(7 rows)</p> | itemName | brandname | price | condition | Chanel Classic Flap Bag Black | Chanel | 3500.00 | Excellent | Celine Luggage Tote Phoebe Era | Celine | 2800.00 | Very Good | Prada Galleria Bag Black | Prada | 1800.00 | Excellent | Gucci Velvet Jacket Size 40 | Gucci | 1500.00 | Very Good | Balenciaga Blazer Size 38 | Balenciaga | 1200.00 | Good | Hermes Silk Scarf Les Chevaux | Hermes | 450.00 | Very Good | Chanel CC Pearl Earrings | Chanel | 380.00 | Excellent |
| itemName                                                                                                                                  | brandname                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | price    | condition |       |           |                               |        |         |           |                                |        |         |           |                          |       |         |           |                             |       |         |           |                           |            |         |      |                               |        |        |           |                          |        |        |           |
| Chanel Classic Flap Bag Black                                                                                                             | Chanel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3500.00  | Excellent |       |           |                               |        |         |           |                                |        |         |           |                          |       |         |           |                             |       |         |           |                           |            |         |      |                               |        |        |           |                          |        |        |           |
| Celine Luggage Tote Phoebe Era                                                                                                            | Celine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2800.00  | Very Good |       |           |                               |        |         |           |                                |        |         |           |                          |       |         |           |                             |       |         |           |                           |            |         |      |                               |        |        |           |                          |        |        |           |
| Prada Galleria Bag Black                                                                                                                  | Prada                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1800.00  | Excellent |       |           |                               |        |         |           |                                |        |         |           |                          |       |         |           |                             |       |         |           |                           |            |         |      |                               |        |        |           |                          |        |        |           |
| Gucci Velvet Jacket Size 40                                                                                                               | Gucci                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1500.00  | Very Good |       |           |                               |        |         |           |                                |        |         |           |                          |       |         |           |                             |       |         |           |                           |            |         |      |                               |        |        |           |                          |        |        |           |
| Balenciaga Blazer Size 38                                                                                                                 | Balenciaga                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1200.00  | Good      |       |           |                               |        |         |           |                                |        |         |           |                          |       |         |           |                             |       |         |           |                           |            |         |      |                               |        |        |           |                          |        |        |           |
| Hermes Silk Scarf Les Chevaux                                                                                                             | Hermes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 450.00   | Very Good |       |           |                               |        |         |           |                                |        |         |           |                          |       |         |           |                             |       |         |           |                           |            |         |      |                               |        |        |           |                          |        |        |           |
| Chanel CC Pearl Earrings                                                                                                                  | Chanel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 380.00   | Excellent |       |           |                               |        |         |           |                                |        |         |           |                          |       |         |           |                             |       |         |           |                           |            |         |      |                               |        |        |           |                          |        |        |           |



# Query 2: Natural Join

**QUERY 2 :** List item name, brand, price and seller name for all items from French brands, ordered by brand name.

| SQL Query                                                                                                                                                                                                       | Query Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |               |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------|-------|------------|--------------------------------|--------|---------|---------------|-------------------------------|--------|---------|---------------|--------------------------|--------|--------|---------------|-----------------------|--------|---------|---------------|-------------------------------|--------|--------|---------------|-------------------------|---------------|---------|--------------|-------------------------------|---------------|--------|--------------|
| <pre>SELECT itemName,       brandName,       price,       sellerName FROM   ReSee_Item natural join       ReSee_Seller natural join       ReSee_Brand WHERE  countryOrigin = 'France' ORDER BY brandName;</pre> | <table><tr><th>itemName</th><th>brandname</th><th>price</th><th>sellername</th></tr><tr><td>Celine Luggage Tote Phoebe Era</td><td>Celine</td><td>2800.00</td><td>Anna Kowalski</td></tr><tr><td>Chanel Classic Flap Bag Black</td><td>Chanel</td><td>3500.00</td><td>Sofia Bernard</td></tr><tr><td>Chanel CC Pearl Earrings</td><td>Chanel</td><td>380.00</td><td>Sofia Bernard</td></tr><tr><td>Hermes Cape Cod Watch</td><td>Hermes</td><td>2200.00</td><td>Anna Kowalski</td></tr><tr><td>Hermes Silk Scarf Les Chevaux</td><td>Hermes</td><td>450.00</td><td>Anna Kowalski</td></tr><tr><td>Louis Vuitton Speedy 30</td><td>Louis Vuitton</td><td>1100.00</td><td>Priya Sharma</td></tr><tr><td>Saint Laurent Sandals Size 37</td><td>Saint Laurent</td><td>750.00</td><td>Giulia Rossi</td></tr></table> <p>(7 rows)</p> | itemName | brandname     | price | sellername | Celine Luggage Tote Phoebe Era | Celine | 2800.00 | Anna Kowalski | Chanel Classic Flap Bag Black | Chanel | 3500.00 | Sofia Bernard | Chanel CC Pearl Earrings | Chanel | 380.00 | Sofia Bernard | Hermes Cape Cod Watch | Hermes | 2200.00 | Anna Kowalski | Hermes Silk Scarf Les Chevaux | Hermes | 450.00 | Anna Kowalski | Louis Vuitton Speedy 30 | Louis Vuitton | 1100.00 | Priya Sharma | Saint Laurent Sandals Size 37 | Saint Laurent | 750.00 | Giulia Rossi |
| itemName                                                                                                                                                                                                        | brandname                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | price    | sellername    |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Celine Luggage Tote Phoebe Era                                                                                                                                                                                  | Celine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2800.00  | Anna Kowalski |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Chanel Classic Flap Bag Black                                                                                                                                                                                   | Chanel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3500.00  | Sofia Bernard |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Chanel CC Pearl Earrings                                                                                                                                                                                        | Chanel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 380.00   | Sofia Bernard |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Hermes Cape Cod Watch                                                                                                                                                                                           | Hermes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2200.00  | Anna Kowalski |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Hermes Silk Scarf Les Chevaux                                                                                                                                                                                   | Hermes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 450.00   | Anna Kowalski |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Louis Vuitton Speedy 30                                                                                                                                                                                         | Louis Vuitton                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1100.00  | Priya Sharma  |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Saint Laurent Sandals Size 37                                                                                                                                                                                   | Saint Laurent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 750.00   | Giulia Rossi  |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |

# Query 3: Cross Product (equivalent to Query 2)

**QUERY 3 :** This re-expresses Query 2 using cross product notation instead of natural join.

| SQL Query                                                                                                                                                                                                                                                                     | Query Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |               |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------|-------|------------|--------------------------------|--------|---------|---------------|-------------------------------|--------|---------|---------------|--------------------------|--------|--------|---------------|-----------------------|--------|---------|---------------|-------------------------------|--------|--------|---------------|-------------------------|---------------|---------|--------------|-------------------------------|---------------|--------|--------------|
| <pre>SELECT i.itemName,        i.brandName,        i.price,        s.sellerName FROM   ReSee_Item i, ReSee_Seller s,        ReSee_Brand b WHERE  i.sellerEmail = s.sellerEmail AND    i.brandName = b.brandName AND    b.countryOrigin = 'France' ORDER BY i.brandName;</pre> | <table><tr><th>itemname</th><th>brandname</th><th>price</th><th>sellername</th></tr><tr><td>Celine Luggage Tote Phoebe Era</td><td>Celine</td><td>2800.00</td><td>Anna Kowalski</td></tr><tr><td>Chanel Classic Flap Bag Black</td><td>Chanel</td><td>3500.00</td><td>Sofia Bernard</td></tr><tr><td>Chanel CC Pearl Earrings</td><td>Chanel</td><td>380.00</td><td>Sofia Bernard</td></tr><tr><td>Hermes Cape Cod Watch</td><td>Hermes</td><td>2200.00</td><td>Anna Kowalski</td></tr><tr><td>Hermes Silk Scarf Les Chevaux</td><td>Hermes</td><td>450.00</td><td>Anna Kowalski</td></tr><tr><td>Louis Vuitton Speedy 30</td><td>Louis Vuitton</td><td>1100.00</td><td>Priya Sharma</td></tr><tr><td>Saint Laurent Sandals Size 37</td><td>Saint Laurent</td><td>750.00</td><td>Giulia Rossi</td></tr></table> <p>(7 rows)</p> | itemname | brandname     | price | sellername | Celine Luggage Tote Phoebe Era | Celine | 2800.00 | Anna Kowalski | Chanel Classic Flap Bag Black | Chanel | 3500.00 | Sofia Bernard | Chanel CC Pearl Earrings | Chanel | 380.00 | Sofia Bernard | Hermes Cape Cod Watch | Hermes | 2200.00 | Anna Kowalski | Hermes Silk Scarf Les Chevaux | Hermes | 450.00 | Anna Kowalski | Louis Vuitton Speedy 30 | Louis Vuitton | 1100.00 | Priya Sharma | Saint Laurent Sandals Size 37 | Saint Laurent | 750.00 | Giulia Rossi |
| itemname                                                                                                                                                                                                                                                                      | brandname                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | price    | sellername    |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Celine Luggage Tote Phoebe Era                                                                                                                                                                                                                                                | Celine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2800.00  | Anna Kowalski |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Chanel Classic Flap Bag Black                                                                                                                                                                                                                                                 | Chanel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3500.00  | Sofia Bernard |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Chanel CC Pearl Earrings                                                                                                                                                                                                                                                      | Chanel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 380.00   | Sofia Bernard |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Hermes Cape Cod Watch                                                                                                                                                                                                                                                         | Hermes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2200.00  | Anna Kowalski |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Hermes Silk Scarf Les Chevaux                                                                                                                                                                                                                                                 | Hermes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 450.00   | Anna Kowalski |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Louis Vuitton Speedy 30                                                                                                                                                                                                                                                       | Louis Vuitton                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1100.00  | Priya Sharma  |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |
| Saint Laurent Sandals Size 37                                                                                                                                                                                                                                                 | Saint Laurent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 750.00   | Giulia Rossi  |       |            |                                |        |         |               |                               |        |         |               |                          |        |        |               |                       |        |         |               |                               |        |        |               |                         |               |         |              |                               |               |        |              |



# Query 4: Group By with Having

**QUERY 4 :** List each brand and the total number of items it has listed. Only show brands that have more than one item listed, ordered by total items from highest to lowest.

| SQL Query                                                                                                                                                              | Query Output                                                                                                                                                                     |           |             |        |   |        |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|--------|---|--------|---|
| <pre>SELECT brandName,<br/>       COUNT(*) AS total_items<br/>FROM   ReSee_Item<br/>GROUP BY brandName<br/>HAVING COUNT(*) &gt; 1<br/>ORDER BY total_items DESC;</pre> | <table><thead><tr><th>brandname</th><th>total_items</th></tr></thead><tbody><tr><td>Hermes</td><td>2</td></tr><tr><td>Chanel</td><td>2</td></tr></tbody></table> <p>(2 rows)</p> | brandname | total_items | Hermes | 2 | Chanel | 2 |
| brandname                                                                                                                                                              | total_items                                                                                                                                                                      |           |             |        |   |        |   |
| Hermes                                                                                                                                                                 | 2                                                                                                                                                                                |           |             |        |   |        |   |
| Chanel                                                                                                                                                                 | 2                                                                                                                                                                                |           |             |        |   |        |   |

# Query 5: Subquery

**QUERY 5 :** List the name, brand and price of the most expensive item currently available for sale.

| SQL Query                                                                                                                                  | Query Output                                                                                                                                                           |          |           |       |                               |        |         |
|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|-------|-------------------------------|--------|---------|
| <pre>SELECT itemName,   brandName,   price FROM ReSee_Item WHERE price = (SELECT MAX(price)   FROM ReSee_Item   WHERE isSold = 'N');</pre> | <table><tr><th>itemname</th><th>brandname</th><th>price</th></tr><tr><td>Chanel Classic Flap Bag Black</td><td>Chanel</td><td>3500.00</td></tr></table> <p>(1 row)</p> | itemname | brandname | price | Chanel Classic Flap Bag Black | Chanel | 3500.00 |
| itemname                                                                                                                                   | brandname                                                                                                                                                              | price    |           |       |                               |        |         |
| Chanel Classic Flap Bag Black                                                                                                              | Chanel                                                                                                                                                                 | 3500.00  |           |       |                               |        |         |



# Query 6: Self Join (Cross Product)

QUERY 6 : Find all orders that contain more than one item by joining ORDER\_ITEM with itself. Uses aliases oi1 and oi2.

| SQL Query                                                                                                                                                                                                                                                                       | Query Output                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <pre>SELECT oi1.orderNo,<br/>       oi1.itemNo  AS first_item,<br/>       oi2.itemNo  AS second_item<br/>FROM   ReSee_Order_Item oi1,<br/>       ReSee_Order_Item oi2<br/>WHERE  oi1.orderNo= oi2.orderNo<br/>AND    oi1.itemNo &lt; oi2.itemNo<br/>ORDER BY oi1.orderNo;</pre> | <pre>orderno   first_item   second_item<br/>-----+-----+-----<br/>      1004            3            8<br/>(1 row)</pre> |

# Domain Integrity — CHECK Constraints

CHECK constraints enforce that only valid data can be inserted into the database.

1

## ReSee\_Item —di\_item\_condition

```
CONSTRAINT di_item_condition CHECK (condition IN  
(  
'Excellent', 'Very Good', 'Good', 'Fair'  
))
```

**Only 4 valid condition values accepted. Any other value (example: 'Average') will be rejected.**

2

## ReSee\_Order\_Table —di\_order\_status

```
CONSTRAINT di_order_status CHECK (status IN (  
'Pending', 'Confirmed', 'Shipped',  
'Delivered', 'Cancelled'))
```

**Order status must be one of 5 valid values, matching the real ReSee workflow.**

3

## ReSee\_Item —di\_item\_price

```
CONSTRAINT di_item_price CHECK (price > 0)
```

**Price must be a positive number. A price of 0 or negative is not allowed.**

4

## Physical Causes

```
CONSTRAINT di_seller_joinedDate CHECK  
(joinedDate >= '2013-01-01')
```

**Join date cannot be before 2013 — the year ReSee was founded.**



# Referential Integrity — ON DELETE Actions

## ON DELETE RESTRICT

### What it means:

If you try to delete a parent record that has child records, PostgreSQL will BLOCK the deletion and return an error.

Example in the database:

```
-- Delete seller Anna Kowalski
DELETE FROM ReSee_Seller
WHERE sellerEmail = 'anna.k@email.com';
```

### Result:

```
ERROR:  update or delete on table "resee_seller" violate
s foreign key constraint "resee_itemfk_seller" on table
"resee_item"
DETAIL:  Key (selleremail)=(anna.k@email.com) is still r
eferenced from table "resee_item".
```

**Anna has 3 items in ITEM — she cannot be deleted until those items are removed first.**

## ON DELETE CASCADE

### What it means:

If you delete a parent record, PostgreSQL will AUTOMATICALLY delete all child records too, no error, the deletion cascades down.

Example in the database:

```
-- Delete order 1004
DELETE FROM ReSee_Orders
WHERE orderNo = 1004;
```

### Result:

```
postgres=# -- Delete order 1004
DELETE FROM ReSee_Orders
WHERE orderNo = 1004;
DELETE 1
```

**Deleting order 1004 automatically removes its two rows in ORDER\_ITEM (items 3 and 8).**

# Database View — AvailableItems

A VIEW is a saved query that acts like a virtual table. Users can query it directly without writing the full SQL each time.

| SQL Query                                                                                                                                                                                                                                                                          | Query Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>CREATE VIEW AvailableItems AS SELECT itemNo, itemName,        brandName, typeName,        price, condition FROM   ReSee_Item WHERE  isSold = 'N' AND    isAuthenticated = 'Y';</pre> <p><b>How to use the VIEW</b></p> <pre><b>SELECT * FROM</b> <b>AvailableItems;</b></pre> | <pre>postgres=# SELECT * FROM AvailableItems;  itemno                itemname                brandname   typeName    price    condition -----+-----+-----+-----+-----+-----       1   Chanel Classic Flap Bag Black        Chanel      Handbag    3500.00   Excellent       2   Hermes Silk Scarf Les Chevaux       Hermes      Scarf        450.00   Very Good       3   Celine Luggage Tote Phoebe Era      Celine      Handbag    2800.00   Very Good       4   Balenciaga Blazer Size 38            Balenciaga   Clothing   1200.00   Good       7   Prada Galleria Bag Black             Prada       Handbag    1800.00   Excellent       8   Chanel CC Pearl Earrings             Chanel      Jewellery     380.00   Excellent      10   Gucci Velvet Jacket Size 40          Gucci       Clothing   1500.00   Very Good (7 rows)</pre> |

## Why use a view?

Simplifies queries, users don't need to remember the WHERE conditions every time.



# THANK YOU FOR YOUR ATTENTION

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